9 (a) (i) The electric field strength at a point is defined as the electric force per unit positive charge placed at that point: B1

(ii) Electron gains speed as it moves from point A of 200V to 300V. B2 for all 3

Its speed remains approximately constant when it is between the region of 300V.

Its speed decreases (to the same initial speed) as it moves from 300V to point B

(iii) Loss in kinetic energy = gain in electric potential energy

$= q\,(V_C - V_A)$

$= (-1.60 \times 10^{-19})\,(0 - 200)$ C1

$= 3.20 \times 10^{-17}$ J A1

To deduct one mark if answer does not reflect negative value which represents the loss in KE.

(iv) Distance between 300V and 100V beside B is 2.7×10^{-2} m

Magnitude of field strength

$= |\Delta V/\Delta x|$

$= (300 - 100)/(2.7\times10^{-2})$ C1

$= 7400$ V m^{-1} (7100 – 7700) A1

(v) Vertically downwards B1

(vi) $F = qE$

$=(1.60\times10^{-19})(7400)$

$=1.18\times10^{-15}$ N A1

Vertically upwards. A1

(b) (i) The gravitational force acting on the object provides the centripetal force.

$$\frac{GMm}{R^2} = mR\omega^2$$ B1

$$\omega = \sqrt{\frac{GM}{R^3}}$$

(ii) A longer arrow acts towards the Sun. B1
A shorter arrow acts towards the Earth

(iii) The centripetal force acting on the SOHO depends on both the gravitational forces due to the Sun and the Earth. Hence, the expression in (i) does not apply to the angular velocity for SOHO. B1

Due to the gravitational force acting on SOHO by the Earth, the net centripetal force acting on SOHO is lower. This will result in a lower angular velocity for SOHO that can be comparable to that of the Earth at a further distance from the Sun. B1

(iv) $$\frac{GMm}{R^2} - \frac{GM_{Earth}m}{R_{Earth}{}^2} = mR\left(\frac{2\pi}{T}\right)^2$$

$$\frac{GM}{R^2} - \frac{GM_{Earth}}{R_{Earth}{}^2} = R\left(\frac{2\pi}{T}\right)^2$$

where m is the mass of SOHO, M_{Earth} is the mass of the Earth and R_{Earth} is the distance from SOHO to the Earth. From the above equation, the orbital period does not depend on the mass of SOHO B1

(c) **(i)** Since $\frac{GM_E}{R_E{}^2} = 9.81$, $R_E = \sqrt{\frac{G(6.0 \text{ x } 10^{24})}{9.81}} = 6.39 \text{ x } 10^6$ m C1

$$v = \sqrt{\frac{G(6.0 \text{ x } 10^{24})}{(7.0+6.39) \text{ x } 10^6}} = 5470 \text{ m s}^{-1}$$ A1

(ii) GPE at infinity + KE at infinity
= GPE of stone on satellite + KE of stone on satellite

$$0 + 0 = -\frac{G(6.0 \text{ x } 10^{24})m}{(7.0+6.39) \text{ x } 10^6} + \frac{mv^2}{2}$$

v = 7731 m s^{-1} C1

Minimum projection speed = 7731 + 5470 = 1.3 x 10^4 m s^{-1} A1

(iii) The lowest speed of projection will be 7731 – 5470 = 2261 m s^{-1} relative to the speed of satellite and the stone should be projected in the same direction as the movement of satellite. Any other direction of projections will not lead to lower speed to escape from the Earth's gravitational field. B1

10 **(a)** **(i)** Magnetic force provides centripetal force M1